Check in #2: Climate and Environment Group 3: Carbon Pricing

Qingyue Chen, Puran Dou, Yi Dou, Gavin Gondalwala, Catherine Morris

Research Question: Does EU ETS carbon pricing cause carbon leakage in Italian chemical manufacturing primarily through upstream supply chain adjustment, measured as increased imports of industrial inputs sourced from unregulated suppliers, rather than through substitution toward imported final goods?

Time Period: 2020-2027

I. Introduction

1.1 Background

The EU operates one of the most comprehensive emissions regulatory frameworks in the world, anchored by the Emissions Trading System (ETS). In operation since 2005, the ETS requires energy-intensive industries to purchase allowances for each tonne of CO_2 emitted, with the total cap on allowances declining each year. The system currently covers heavy industry, power generation, aviation, and maritime transport, and explicitly includes the production of bulk organic and inorganic chemicals — the sector at the center of this study.

While the ETS has reduced covered-sector emissions by approximately 50% since its inception, it has faced significant and growing political opposition. Italy, leading an eleven-member coalition of member states known as the "Friends of Industry," has called for a temporary suspension of the ETS pending comprehensive reform, arguing that the system functions as a *de facto* tax on European industry and undermines international competitiveness. This position is contested within the EU: eight member states, including Denmark, Sweden, and the Netherlands, have formally opposed suspension, describing the ETS as the cornerstone of EU climate policy. Others argue that the true culprit behind prohibitive prices is the EU's continued dependence on fossil fuels—an issue exacerbated by the war in Iran.

While recent events have heightened this debate, at a fundamental level, it reflects a genuine tension between decarbonization objectives and industrial competitiveness that remains unresolved at the EU level. As a European Commission report argues, the expansion of environmental and chemical regulations may outpace firms' ability to adapt—which could weaken their economic competitiveness, and force them to cede market share to firms in countries with less stringent regulations. EU data show that overall, greenhouse gas emissions from the chemical industry decreased by 9 percent between 2012 and 2021. However, much of that reduction occurred before 2015, and since then, emissions have been stable, indicating that progress on reducing emissions overall has stagnated. In comparison, Italian firms have made significant progress, reducing GHG emissions by 59% since 1990.

The latter point connects to a central concern in this debate: carbon leakage. If ETS compliance costs cause firms to shift sourcing or production toward suppliers in countries with less stringent emissions regulation, the goal of reducing global emissions is simply outsourced to another country, negating efforts to reduce global emissions overall. This study proposes to

examine that mechanism directly, focusing on the Italian chemical sector's sourcing of intermediate goods during the period 2020–27. In order to study the impact of ETS on firm behavior, we propose to introduce a voucher to firms in the chemical sector, as a temporary, targeted subsidy on compliance costs. The cost will be absorbed by the Italian government.

1.2 Motivation

The Italian chemical sector provides an unusually clean setting in which to study the relationship between ETS compliance costs and intermediate goods sourcing. Italy is the third-largest chemical producer in Europe, and the 10th largest worldwide. The broader chemicals sector — classified under HS Chapters 28 and 29 — is directly covered by ETS obligations on the production side. Critically, the sector relies heavily on upstream chemical intermediates that are not sourced domestically: European producers depend on China for an estimated 75–85% of the intermediates used in Active Pharmaceutical Ingredient (API) manufacturing, for example.

Italian leaders seeking a suspension to the ETS argue that the program has had a damaging effect on the sector's competitiveness. For example, Italian Industry minister Adolfo Urso has said, "If we are in the face of the collapse of the European chemical industry and the crisis of European ideology, we cannot wait for the time of negotiations within the European Union to find a solution."

This configuration creates a testable mechanism. ETS compliance costs raise the effective cost of domestic production, potentially making imported intermediates from lower-cost, higher-emissions suppliers relatively more attractive. If firms respond to ETS cost pressure by shifting sourcing toward unregulated suppliers, the environmental benefit of the regulation is partially or fully offset by upstream emissions abroad. Conversely, if ETS cost relief — modeled here as an emissions credit — induces firms to shift sourcing back toward EU-based suppliers, the intervention would provide evidence that carbon pricing meaningfully shapes global supply chain composition.

Existing literature on carbon leakage has focused primarily on production relocation and sectoral output, with less attention to intermediate goods sourcing as a margin of adjustment. This study addresses that gap. By randomizing an emissions credit across eligible Italian chemical firms and tracking changes in sourcing geography and composition, we aim to identify the causal effect of ETS compliance costs on firms' procurement decisions — and, by extension, on the net emissions consequences of carbon pricing policy. The results are directly relevant to the ongoing EU debate over ETS reform, providing micro-level evidence on a mechanism that has been theorized but not cleanly identified.

II. Experimental Design

2.1 Intervention/Manipulation

The intervention is a temporary emissions credit provided to eligible Italian chemical firms. The credit is designed to reduce the effective compliance costs associated with the EU ETS, thereby lowering the carbon-cost burden embedded in firms' production and sourcing decisions. The purpose of the intervention is to test whether reducing ETS-related costs induces changes in firms' sourcing of intermediate chemical goods, particularly whether firms shift procurement across suppliers located in EU ETS and unregulated suppliers.

2.2 Enrollment Criteria/Sampling Strategy

The study population consists of firms in the Italian chemical sector, specifically those that produce or import products classified under HS Chapters 28 and 29. To be included in the sample, firms must meet the following criteria: first, they must have observable purchasing or import activity in intermediate chemical goods during the baseline period; second, they must remain active during the study period so that pre- and post-intervention comparisons can be made; and third, they must have a sufficient baseline scale of procurement to allow the study to detect changes in sourcing patterns and procurement volume before and after the intervention. Federchimica Confindustria categorizes chemical firms into three broad groups, which will be stratified in the experiment:

1. Foreign-owned companies (accounting for 37% of total production value)
2. Medium and large-sized Italian groups (accounting 23% of total production value, and defined as those whose worldwide sales exceed 100 million euros)
3. Italian SMEs (accounting for 40% of total production value)

2.3 Assignment Protocol/Identification Strategy

After baseline data collection, eligible firms will be randomly assigned to either a treatment group that receives the emissions credit or a control group that does not. Random assignment ensures that, on average, treated and control firms are comparable in both observed and unobserved characteristics prior to the intervention. The causal effect of the emissions credit will be identified by comparing post-intervention sourcing outcomes between the two groups. The main outcomes include the share of intermediate imports sourced from unregulated countries, the ratio of imports from unregulated suppliers to EU-ETS countries, and the total volume of imported intermediate chemical goods. If the experiment is properly randomized, the post-treatment difference between treated and control firms can be interpreted as the causal effect of reducing effective ETS compliance costs.

III. Data

3.1 Inputs, outputs and outcomes of interest

In this study, we focus on the Italian chemical sector, defined as products classified under HS Chapters 28 and 29 (inorganic and organic chemicals). We conceptualize firm behavior

within a production and sourcing framework, where firms combine intermediate inputs to produce output under changing carbon cost conditions.

Inputs are proxied by imported intermediate chemical goods, measured using trade data at the HS6 level for HS28–29. We restrict the sample to Italian imports and interpret these flows as intermediate inputs into domestic chemical production. Input measures include total import value as well as the distribution of imports across these partner groups. To capture sourcing patterns, we classify partner countries into two groups: EU-ETS-Covered Suppliers and Unregulated Suppliers:

EU-ETS-Covered Suppliers	Unregulated Suppliers
All jurisdictions are covered by EU ETS for the chemical sector	*Jurisdictions where there is no national ETS, or where the chemical sector is excluded*
EU Countries	Such As: China India United States

Outputs refer to production-related outcomes in the Italian chemical sector. Given the absence of firm-level production data, we proxy output using export values of HS28–29 chemical products from Italy, which reflect changes in production scale and international competitiveness. This allows us to assess whether changes in input costs and sourcing patterns translate into observable shifts in sectoral output.

Outcomes of interest capture changes in sourcing behavior consistent with carbon leakage. The main outcome variables include:

(1) the share of imports sourced from unregulated suppliers,
(2) the ratio of imports from unregulated suppliers to EU-ETS countries, and
(3) the total volume of imported chemical inputs.

3.2 Data Collection Strategy & Data Sources

There will be a baseline survey issued to firms at the beginning of the study which collects information on their current sourcing partners, with an endline survey at the end to assess behavior changes. During the baseline, firms will be asked questions which build on the following:

1. What is the quantity of Schedule 28 and 29 chemicals your firm has sourced domestically (within Italy) in the last fiscal quarter for use in your manufacturing process?
2. What is the quantity of Schedule 28 and 29 chemicals your firm has imported in the last fiscal quarter for use in your manufacturing process?
3. What country were those chemicals sourced from?

4. Why did you choose this sourcing partner? *(Firms will select primary reasons for choosing their sourcing partner from a categorical list (e.g., Price, Material Quality, Existing Long-Term Contract, Geographic Proximity) to allow for fixed-effects controls in future studies.)*
5. What were the estimated EU Carbon Border Adjustment Mechanism (CBAM) certificate costs associated with these imports?
6. What was your facility's total ETS Allowance (EUA) surrender for domestic manufacturing operations during the last fiscal year? *(The European Union defines 1 allowance as the right to emit 1 tonne of CO2 equivalent [European Commission, n.d.])*.
7. Are final goods intended for domestic markets, export abroad, or some mix?

Following the survey, the study team will append national emissions regulations in source countries and a carbon equivalent calculation to survey output.

Identifying quantities and origins allows the study to track changes in sourcing behavior and quantities to provide context for the environmental and economic impacts of ETS. At the conclusion of the study, firms will be issued a similar endline survey with the addition of calculations for their total imports over the course of the study. Once the final data is appended, the study team will investigate if sourcing has shifted in a statistically significant manner.

3.3 Hypotheses and Outcomes

The study's working hypothesis is that by working with Italian regional governments to secure EUA vouchers and provisioning them to firms in the treatment group, the effective marginal cost of domestic and EU-based production will be reduced–ultimately incentivizing firms to substitute away from carbon-intensive, non-ETS sourcing. To evaluate this causal mechanism, the study will test the following hypotheses:

<u>Outcome 1: Sourcing Composition/Leakage Reversal</u>

H0: The provision of an emissions credit has no effect on the share of intermediate chemical goods sourced from unregulated suppliers.

H1: Firms receiving the credit will significantly decrease the share of their total intermediate imports sourced from unregulated suppliers compared to the control group.

Data construction: Calculated as total import value from unregulated suppliers divided by total import value from all partner countries.

<u>Outcome 2: Sourcing Ratio</u>

H0: The provision of an emissions credit does not alter the ratio of imports from unregulated suppliers to EU-ETS countries.

H1: Firms receiving the credit will exhibit a lower ratio of unregulated suppliers to EU-ETS imports compared to the control group.

Data Construction: Calculated as the total import volume from unregulated suppliers divided by the total import value from EU-ETS countries.

Outcome 3: Total Import Volume

H0: The provision of an emissions credit has no effect on the total volume of imported chemical inputs.

H1: Firms receiving the credit will decrease their total volume of imported intermediate goods as they substitute towards domestic Italian suppliers.

Data construction: Calculated as the aggregate volume of all Schedule 28 and 29 substances imported by the firm over the study period.

While aggregate macroeconomic shocks — such as rising global energy costs — may impact the sector's overall sourcing behavior, the randomized control design ensures these shocks are distributed symmetrically across the treatment and control groups, protecting the internal validity of the Average Treatment Effect (ATE).

IV. Power Calculations (Setup)

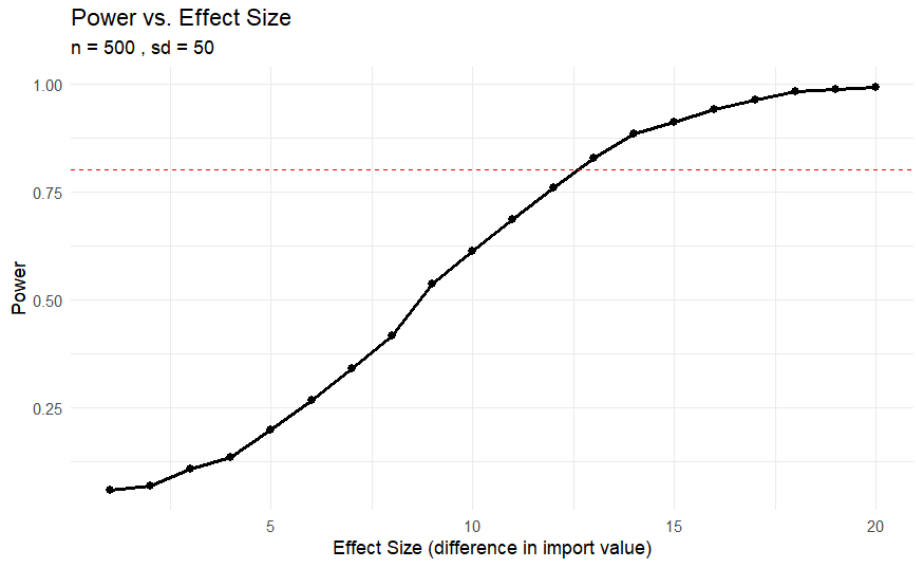
Contextual data is mainly sourced from Volza's Italy Import Trade Data dataset, with additional context provided from the Bank of Italy, the OECD, and the UN Comtrade Database. Firm anonymity is a data constraint which we cannot circumvent; however, to approximate the data, we can use parameters from aggregate level data from the sources mentioned above to simulate the firm level data.

To use these simulated data to conduct power calculations, we have to make some assumptions:

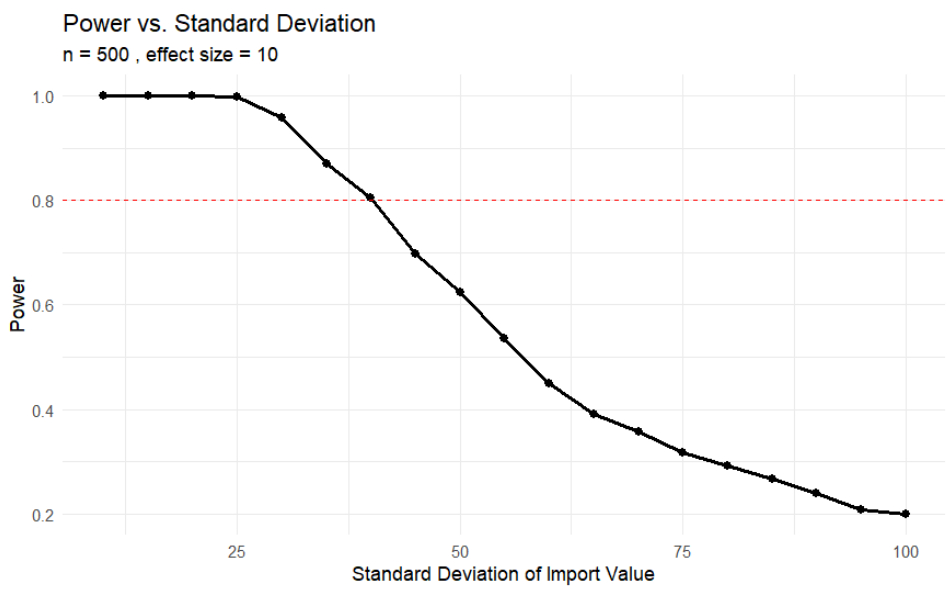
- Normality: The outcome variable is assumed to follow a normal distribution. In practice, firm-level import values are often right-skewed or contain many zeros, which would violate this assumption.
- Homoscedasticity: The variance of the outcome is assumed to be equal across the treatment and control groups.
- Independence: Observations are assumed to be independent of one another. If firms within the same industry or region are correlated, this assumption would be violated, and the actual required sample size would be larger.
- Perfect randomization with no covariates: Treatment assignment is assumed to be perfectly random, and no control variables are included in the model. In practice, controlling for covariates such as firm size or sector can reduce residual variance and improve power.

Besides these assumptions, we also set the significance level and the power threshold at the traditional alpha = 0.05 and beta = 0.2 level. Based on these assumptions and settings, we

calculated the power of our analysis with different values of effect size, standard deviation and sample size. The simulation times for each point are 2,000.

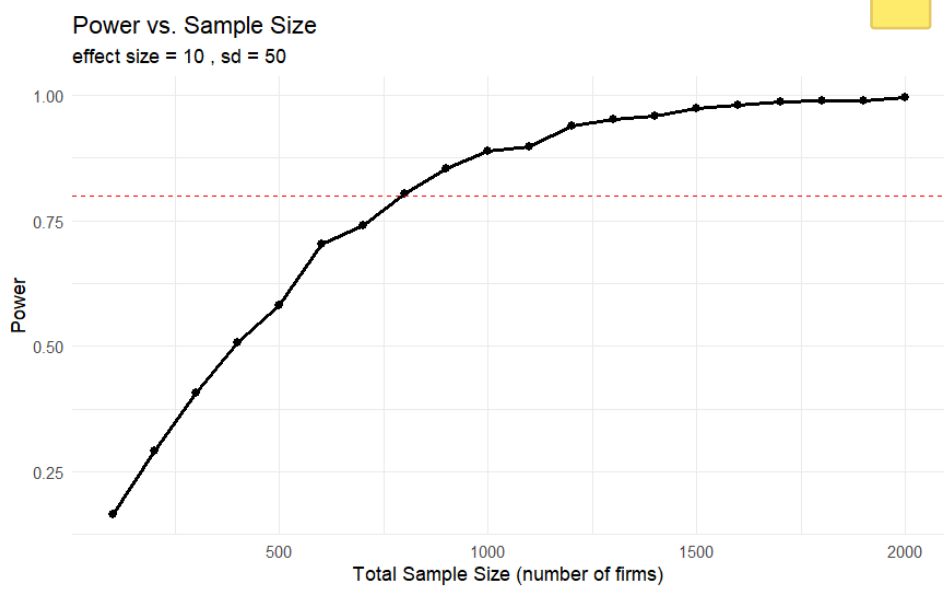


This figure illustrates how statistical power increases as the effect size grows, with the sample size fixed at 500 and the standard deviation at 50. When the effect size is small (below 5), power remains under 0.2, meaning the experiment would rarely detect such a subtle policy impact. Power crosses the conventional 0.8 threshold at an effect size of around 13 to 14, suggesting that with 500 firms, the experiment is well-powered only if the policy is expected to shift import values by at least that amount. This underscores the need for a realistic prior estimate of how large the policy effect is likely to be — if the true effect is small, a much larger sample would be required.



This figure shows how statistical power decreases as the standard deviation of the outcome variable increases, holding the sample size at 500 and the effect size at 10. When the

standard deviation is small (below 30), power is near 1.0, meaning the experiment would almost certainly detect the treatment effect. However, as the standard deviation rises beyond approximately 40, power drops below the 0.8 threshold, indicating that the noise in the data begins to overwhelm the signal. This highlights the importance of having a reasonable estimate of outcome variability when designing an experiment — if firm-level import values are highly dispersed, a larger sample size would be needed to maintain adequate power.



This figure shows how statistical power increases with the total number of firms in the experiment, holding the effect size at 10 and the standard deviation at 50. Power rises steeply at smaller sample sizes and crosses the 0.8 threshold at roughly 800 to 900 firms. Beyond around 1,200 firms, the curve flattens and power approaches 1.0, meaning that additional observations yield diminishing returns. This plot is arguably the most directly useful of the three, as it answers the core design question: given plausible assumptions about the effect size and outcome variability, how many firms need to be recruited to ensure a well-powered experiment.

At this setup stage, these simulations are just illustrative examples intended to demonstrate the power calculation methodology and the general relationship between power and key design parameters. The specific values used for effect size, standard deviation, and sample size are provisional and do not represent the actual power of a real experiment. In the final report, these parameters — particularly the effect size and standard deviation — will be calibrated using existing comparable datasets and justified with empirical evidence.

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